



Time =40 minutes

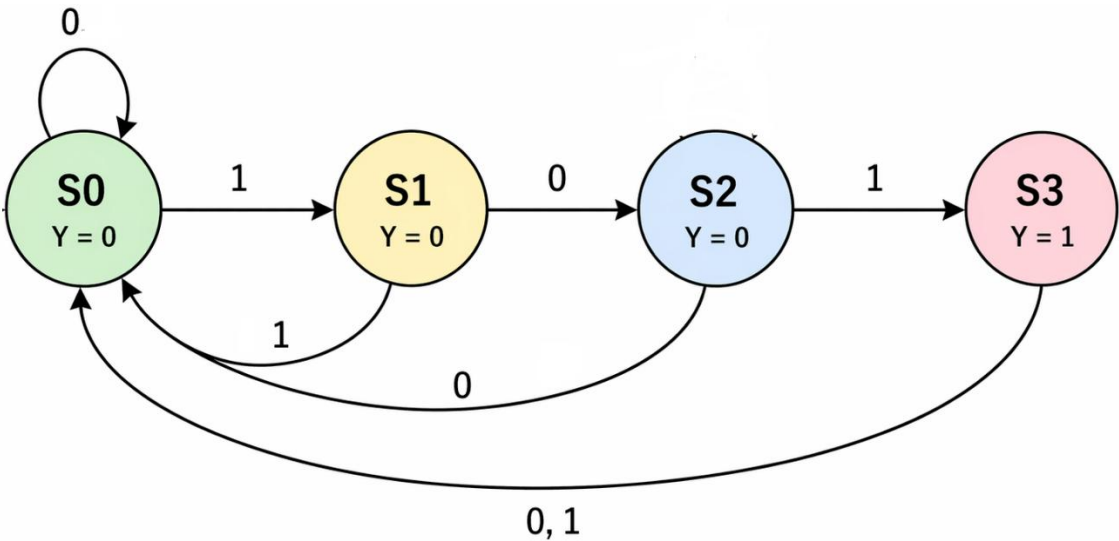
Quiz 06 (CLO 4, Cog 5)

Max Points: 20

Question 1 [3+1+1 points]: A smart door system monitors an input signal X and output a signal Y:

- The system unlocks (Y = 1) only when the sequence 1 → 0 → 1 is detected.
 - The output depends only on successful detection of the sequence.
 - If at any point an incorrect input is received, the system resets to the initial state.
 - After detecting the sequence, the system resets to the initial state on the next input.
- a. Draw the state transition diagram.
- b. Identify whether it is a Mealy or Moore machine with justification.
- c. State the minimum number of flip-flops required.

a)



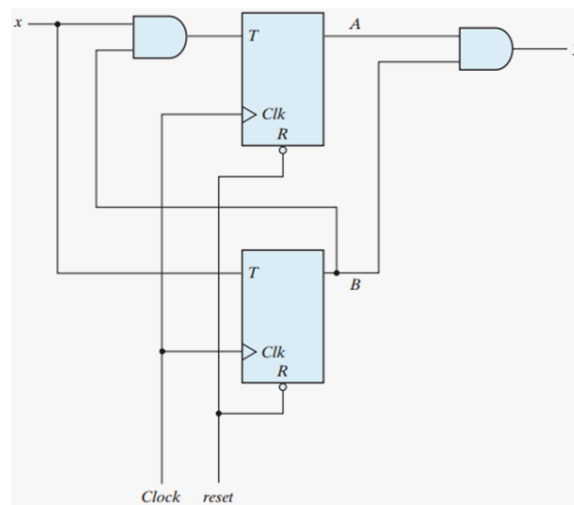
- b) Moore Machine because output depends only on the current state.
- c) 2 flipflops

Question 2 [2 points] A sequential element is described by the following behavior:

Present State	Next State	State
0	0	Memory/ reset
X	1	Set
X	0	Reset
1	1	Memory/set

Classify each transition as Memory, Reset, Set.

Question 3 [9 points]: Implement the given sequential circuit using D flip-flops. Derive the required input equations for the D flip-flops, and draw the final logic circuit.



Hint: write input and output equations of the given circuit, draw state transition table and derive new equations for the D-FFs.

$$T_A = xB, \quad T_B = x, \quad y = AB$$

You can use the following table for working if need be.

A	B	x	A(t+1)	B(t+1)	Y	Ta	Tb
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	1
0	1	0	0	1	0	0	0
0	1	1	1	0	0	1	1
1	0	0	1	0	0	0	0
1	0	1	1	1	0	0	1
1	1	0	1	1	1	0	0
1	1	1	0	0	1	1	1

$$D_A = \sum (3,4,5,6)$$

$$D_B = \sum (1,2,5,6)$$

$$y = \sum (6,7)$$

Free to use K-Maps, decoder or Mux to implement the circuit

Question 4 [4 points]: Complete the timing diagram for the given circuit by drawing the waveforms for signals Y and Q. Based on your result, determine whether the flip-flop is rising-edge triggered or falling-edge triggered.

